EMI-Net: An End-to-End Mechanism-Driven Interpretable Network for SAR Target Recognition Under EOCs

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***Abstract*— Most existing synthetic aperture radar (SAR) target recognition methods based on deep learning are of black box structure and data-drive networks, which are faced with the issue of severe performance degradation under extended operating conditions (EOCs). To address the issue, this article proposes an end-to-end mechanism-driven interpretable network (EMI-Net) for SAR target recognition under EOCs. The EMI-Net achieves the integration of scattering center feature extraction and target recognition in an end-to-end framework to avoid the mismatch between scattering center features with the classifier and also explore the representative electromagnetic characteristics that are useful for recognition under EOCs. In EMI-Net, by unfolding a sparse-solving algorithm and integrating the scattering center model into deep networks, our model contains feature encoding and decoding procedures of SAR images. Thus, EMI-Net is an interpretable deep unfolding network that is driven by physical mechanisms to precisely learn scattering center features reflecting locations and amplitude of SAR targets based on the deep learning mode. Our EMI-Net divides input images into multiple patches based on the image-domain scattering center model to reduce its computation and space complexities. In addition, EMI-Net treats the scattering centers as discrete 3-D point data and designs a point cloud network (PCN) as a classifier to explore permutation-invariant representations for recognition. Results on the measured dataset validate that EMI-Net gains superior scattering center extraction and target recognition results under EOCs, and also shows high time-efficiency and low memory requirement.**

***Index Terms*— Interpretability, point cloud network (PCN), scattering center extraction, synthetic aperture radar (SAR), target recognition.**

I. Introduction

As A moving radar system, synthetic aperture radar (SAR) is widely utilized in various cases of all-time and all-weather. With the SAR imaging system, a large number of high-resolution SAR images can be collected, which is devoted to the rapid development of SAR automatic target recognition (ATR). In [1], [2], and [3], the Lincoln laboratory defines that the SAR ATR task mainly contains three steps: detection, discrimination, and recognition. The target detection locates the target regions of interest and obtains candidate targets, which are fed to target discrimination to remove clutters from candidate targets and collect target chips, and then target recognition focuses on deciding specific categories of target chips. As an important stage in the SAR ATR system, target recognition has aroused widespread concern in remote sensing image processing fields.

We will present a brief review of the development of SAR target recognition methods. In the work [4], [5], [6], some traditional geometric features of SAR images are first extracted, including Krawtchouk moment features, kurtosis wavelet energy features, and Zernike moment features, and then various classifiers are designed for SAR target recognition. Moreover, Zhao and Principe [7] utilize principal component analysis (PCA) to extract subspace features with several orthogonal principal components, and then adopt the linear support vector machine (LSVM) for target recognition. The K-singularly valuable decomposition (K-SVD) [8] and linear discriminant analysis (LDA) [9] are respectively used to extract the over-complete features and low-dimensional features, which are then fed to the LSVM for recognition. In addition, Ding et al. [10] first extract scattering center features via approximate maximum likelihood (AML) algorithm [51], and then they develop a template matching method based on similarity measure for SAR target recognition. Moreover, the sparse solving algorithms, such as orthogonal matching pursuit (OMP) [11], relaxation (RELAX) [36], Bayesian compressive sensing (BCS) [12], approximate message passing (AMP) [27], image-domain sparse representation algorithm [13], and so on, can be utilized for scattering center extraction, and then the scattering centers are used for recognition.

Owing to the powerful feature exploration capability of neural networks, researchers have already put forward lots of deep methods for SAR target recognition. In [14], a full convolution network named A-ConvNet is developed to make use of multiple convolution layers to achieve promising SAR target recognition results. In [15], the backbone structures like VGG16 and VGG19 are utilized to realize target recognition on SAR images. Falqueto et al. [16] develop some data augmentation techniques, including adding noise, image translation, and pose synthesis, to achieve robust results on SAR target recognition. To fuse multiscale features from different

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convolution layers and make full use of the scattering centers of SAR images, Li et al. [17] develop a multiscale CNN based on component analysis (CA-MCNN) to fuse the local and global information of original SAR images, which has gained superior recognition performance on measured SAR images.

Although the methods in [14], [15], [16], and [17] have made a breakthrough in SAR target recognition, they assume that training and test data are obtained in similar operating conditions, i.e., standard operating condition (SOC). When training and test SAR data are collected in different OCs, it is called extended operating condition (EOC) [10]. Generally, EOCs mainly include different versions of types, different resolutions, different levels of noise, occlusion interference, and so on. If the versions of types, resolutions, noise, and occlusion are different between training data with test data, the model learned with training data will no longer be suited for test data. Thus, the recognition performance of existing pure data-driven methods for SOC will degrade under EOCs.

To address the issue of performance degradation under EOCs, the electromagnetic scattering feature-based methods [21], [22], [23], [47] have been proposed to achieve robust SAR target recognition results. Different from existing CNN-based methods focusing on learning structural features, scattering center-based methods focus on utilizing scattering center features of complex images based on SAR imaging theory [19], [20], which contain some rich physically relevant descriptions of targets. Ding et al. [21] propose a fusing global and local features (FGL), where the global subspace features and local scattering center features are first extracted, then the sparse representation-based algorithm and matching-based algorithm are respectively adopted to obtain corresponding recognition results, finally the decision fusion is utilized to attain final results. Zhang et al. [22] develop a novel feature fusion framework named FEC to fuse scattering center features from the bag of visual words algorithm and the deep features from CNNs, and then the fused features by concatenation are fed into a CNN classifier for SAR target recognition. In [23], a decision fusion framework including two branches is built, in which one branch feeds extracted scattering centers into the part convolution network to obtain the local classification results, and another branch feeds images into CNNs to obtain global classification results. Moreover, Liu et al. [47] also develop a feature fusion method for SAR target recognition, which fuses the multilevel scattering features from a point cloud network (PCN) and deep features from CNNs. Based on a measured SAR image dataset, the methods in [21], [22], [23], and [47] have shown promising recognition results under EOCs.

Although the above methods in [21], [22], [23], and [47] have achieved robust recognition performance under EOCs, there are several issues that restrict recognition performance and practicability. First, in [21], [22], [23], and [47], the two procedures of scattering center extraction and classification are independent, i.e., scattering centers are first extracted, and then a classifier is designed and trained with extracted scattering centers. In this way, some scattering centers that are useful for recognition may not be extracted, which restricts recognition results. Second, scattering centers are usually extracted via the sparse optimization algorithms [11], [12], [13], [27], [36], whose prior parameters or iteration termination threshold in methods are experiential values and are hard to determine that impact scattering center extraction effect and further recognition results. Third, since traditional scattering center extraction methods [11], [12], [13], [27], [36] are iterative solving algorithms that consume lots of time, this solving procedure is high time-consuming that restricts their real-time practicability.

As introduced in [48], [49], and [50], the unfolding algorithm is a popular technique to design interpretable network architecture that has received wide theoretical research and practical applications. In this technique, a sparse solving algorithm is first established, and then the algorithm is unfolded by adopting several network blocks to replace each iteration calculation. In each unfolded network block, the hyperparameters of algorithm are updated in a deep learning manner that avoids difficulties of determining prior parameters or iteration termination threshold in traditional algorithms. In this way, a systematic relationship and detailed derivation between an iterative solving algorithm and networks are established by algorithm unfolding. Thus, it is a meaningful way to construct networks with high interpretability, and there are recent survey discussing physics-driven methods [52], [53]. In the articles [52] and [53], the authors review some recent work about deep-unrolling methods that can be classified as two streams, including physics-free [54], [55] and physical-inspired methods [56], [57], such as CSNet [54], ADMM-Net [55], MADUN [56], CASNet [57], and so on. Moreover, the articles [52], [53] further summarize the ideas of constructing deep-unrolling networks, and then compare the characteristics of different methods. Especially, incorporating physical characteristics of SAR system, the physical layers are designed for hybrid modeling to achieving SAR image understanding. Moreover, in [29] and [48], deep-unfolding networks, such as AMP-Net [29], have been developed for compressive image sensing tasks. Inspired by this, in this article, we build an end-to-end mechanism-driven interpretable network (EMI-Net) by unfolding algorithm for SAR target recognition, which can comprehensively address issues including restricted scattering center extraction and recognition performance, poor real-time performance in existing scattering center-based SAR target recognition methods. First of all, different from existing methods with independent two stages, EMI-Net avoids the mismatch between scattering center features with classifier, and learns the scattering centers that are useful for recognition. By unfolding a sparse algorithm and embedding scattering center model that depicts the generative process of radar echoes into deep networks, the EMI-Net contains the feature encoding and decoding procedures to learn the scattering centers containing locations and amplitude information of SAR targets. Based on a deep unfolding manner for sparse encoding of SAR images, EMI-Net can precisely learn scattering centers. Compared with traditional algorithms adopting scattering center model in frequency domain, our model is constructed based on scattering center model in image domain. Thus, the input SAR images are divided into many patches to reduce the dimension of input data, then

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the computation and space complexities of our model are further reduced. Moreover, inspired by the work in article [47], instead of treating scattering centers as regular image structure formats, EMI-Net regards them as discrete 3-D data consisting of 2-D locations (x- and y-axis) and amplitude and thus designs a PCN as a classifier. Compared with existing SAR target recognition methods with low time-efficiency and pure data-driven mode, we summarize three contributions of this article as follows.

1) A novel end-to-end model is developed to achieve the integration of scattering center extraction and recognition in a deep framework. Our model can avoid the mismatch between scattering centers and classifier, and extract scattering centers that contain much representative information of SAR targets and are beneficial for recognition. Experiments reveal that our model learns the scattering centers depicting SAR targets more precisely and improves 2.7% rates than two-stage methods.
2) Since the unfolding algorithm is a typical way to improve the interpretability of networks, our model is a mechanism-driven interpretable model by unfolding a sparse-solving algorithm for scattering center feature extraction based on the physical model of SAR images. Compared with AMP-Net being a real-valued network, EMI-Net is a complex-valued network for complex data processing of complex SAR images, and EMI-Net directly utilizes existing complex stochastic gradient descent algorithms to optimize complex network parameters.
3) A high time-efficiency and low memory requirement way is introduced in our model. Based on the scattering center model in the image domain, our method partitions input images into multiple patches by sliding windows that utilize the original image information for edge padding to eliminate the problem of blocking artifacts caused by dividing patches. Correspondingly, our EMI-Net divides the Fourier base in the image domain into patches for dictionary construction, which differs from the operation in AMP-Net which directly uses a sampling matrix as a dictionary. Experiments show that the computation and space requirement of our model decreased by approximately two orders of magnitude than those of related methods.

We will give the section arrangement of the whole article. In Section II, a brief introduction to the scattering center model of SAR images is presented. And then, in Section III, the details about our EMI-Net will be shown. Then, Section IV gives abundant experimental results on the measured SAR image dataset and illustrates the performance of our method under both SOC and EOCs. Section V concludes the whole article.

II. Scattering Center Model

In the optical zone, the SAR target radar echoes are formed by several isolated 2-D scattering centers on the imaging plane [30], and the echo from scattering centers can be seemed as a function of frequency f and aspect angle ϕ, where $f \in (f_c - B/2, f_c + B/2)$ with f_c denoting the center frequency, B denoting bandwidth, and $\phi \in (-\phi_m/2, \phi_m/2)$ with ϕ_m representing the rotation angle. Thus, the target signal is modeled as the sum of individual scattering centers that constitutes the total backscattered field

$$E(f,\phi) = \sum_{i=1}^{K_0} z_i(f,\phi) \exp\left\{-j\frac{4\pi f}{c}(x_i\cos\phi + y_i\sin\phi)\right\} \tag{1}$$

where $E(f,\phi)$ denotes the measurement matrix, K_0 denotes the number of scattering centers, $j = \sqrt{-1}$ represents the imaginary unit, $z_i(f,\phi)$ denotes the scattering coefficient of ith scattering center, (x_i, y_i) means the location coordinate of ith scattering center, and c is the propagation velocity.

Considering that the noise exists in practical application, we have rewritten the signal model in (1) as a vector-matrix expression

$$\tilde{\mathbf{s}}(f,\phi) = \tilde{\mathbf{D}}(f,\phi)\mathbf{z}(f,\phi) + \tilde{\boldsymbol{\eta}} \tag{2}$$

where $\tilde{\mathbf{s}}(f,\phi)$ denotes the vectorization of the measurement $E(f,\phi)$, $\mathbf{z}(f,\phi)$ represents the scattering coefficient vector, and $\tilde{\boldsymbol{\eta}}$ is the noise. In (2), $\tilde{\mathbf{D}}(f,\phi)$ means the parameterized dictionary in signal domain, each column of which represents the vectorization of measurements, which can be expressed as

$$\tilde{\mathbf{D}}(f,\phi) = \left[\tilde{\mathbf{d}}_1(f,\phi), \ldots, \tilde{\mathbf{d}}_i(f,\phi), \ldots, \tilde{\mathbf{d}}_{K_0}(f,\phi)\right] \tag{3}$$

$$\tilde{\mathbf{d}}_i(f,\phi) = \mathrm{vec}\left(\exp\left[-j\frac{4\pi f}{c}(x_i\cos\phi + y_i\sin\phi)\right]\right) \tag{4}$$

with $\mathrm{vec}(\cdot)$ denoting the vectorization operation.

To achieve scattering center extraction from (2) based on sparse representation theory [11], [31], we need to solve a sparse optimization problem as follows:

$$\begin{aligned} &\arg\min_{\mathbf{z}} \left\|\tilde{\mathbf{s}}(f,\phi) - \tilde{\mathbf{D}}(f,\phi)\mathbf{z}(f,\phi)\right\|_2^2 \\ &\text{s.t. } \|\mathbf{z}(f,\phi)\|_0 = K_0 \end{aligned} \tag{5}$$

where $\|\cdot\|_2$ and $\|\cdot\|_0$ respectively denote l_2 and l_0 norms, and $\|\mathbf{z}(f,\phi)\|_0 = K_0$ means that there are K_0 nonzero elements in $\mathbf{z}(f,\phi)$. Generally, it is hard to obtain the accurate number of scattering centers K_0, but the noise energy of signals can be estimated [32]. Therefore, it is reasonable to terminate iteration when the residual signal energy is lower than the noise energy, and (5) can be transformed as

$$\begin{aligned} &\arg\min_{\mathbf{z}} \left\|\tilde{\mathbf{s}}(f,\phi) - \tilde{\mathbf{D}}(f,\phi)\mathbf{z}(f,\phi)\right\|_2^2 \\ &\text{s.t. } \left\|\tilde{\mathbf{s}}(f,\phi) - \tilde{\mathbf{D}}(f,\phi)\mathbf{z}(f,\phi)\right\|_2^2 < \delta \end{aligned} \tag{6}$$

where δ denotes the terminate iteration threshold, and $\|\tilde{\mathbf{s}}(f,\phi) - \tilde{\mathbf{D}}(f,\phi)\mathbf{z}(f,\phi)\|_2^2$ means the residual signal energy. Equation (6) represents the sparse optimization problem in signal domain, which can also be transformed into image domain, as given by

$$\begin{aligned} &\arg\min_{\mathbf{z}} \|\mathbf{s}(f,\phi) - \mathbf{D}(f,\phi)\mathbf{z}(f,\phi)\|_2^2 \\ &\text{s.t. } \|\mathbf{s}(f,\phi) - \mathbf{D}(f,\phi)\mathbf{z}(f,\phi)\|_2^2 < \delta \end{aligned} \tag{7}$$

where $\mathbf{s}(f,\phi)$ denotes the SAR image, and $\mathbf{D}(f,\phi)$ is obtained from $\tilde{\mathbf{D}}(f,\phi)$ by back projection (BP) imaging algorithm [33].

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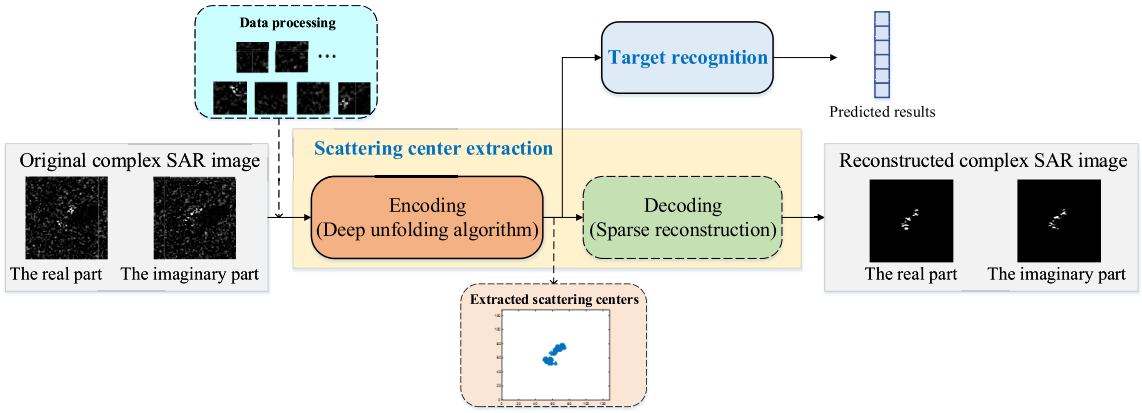


Fig. 1. Overall framework of EMI-Net, where the dashed line represents the network with learnable parameters and solid line denotes the network with fixed parameter.

In the following, we respectively use $\mathbf{D}$, $\mathbf{s}$, and $\mathbf{z}$ to replace $\mathbf{D}(f,\phi)$, $\mathbf{s}(f,\phi)$, and $\mathbf{z}(f,\phi)$ for simplicity, where $\mathbf{D}$ represents the Fourier dictionary in the image domain.

To solve the sparse representation problem in (7), the OMP [11], BCS [12], AMP [27], and the sparse representation algorithm [13] can be used to access the optimal solution for the scattering coefficient $\mathbf{z}(f,\phi)$. However, in (7), due to the high dimension of dictionary $\mathbf{D}$, scattering center extraction process of the methods [11], [12], [13] is usually high time-consuming.

In (7), to reduce the dimension of Fourier dictionary and thus reduce the computation complexity, a way is adopted by dividing the original image $\mathbf{s}$ into m patches representing as $\sigma(\mathbf{s},m)$. Moreover, we adopt the way of sliding windows to obtain image patches by utilizing original image information as edge fills, which can eliminate the problem of blocking artifacts caused by dividing patches. Thus, the (7) can be further rewritten as

$$\begin{aligned} &\arg\min_{\mathbf{z}} \|\sigma(\mathbf{s},m)-\mathbf{D}\mathbf{z}\|_2^2 \\ &\text{s.t. } \|\sigma(\mathbf{s},m)-\mathbf{D}\mathbf{z}\|_2^2 < \delta \end{aligned} \tag{8}$$

where $\mathbf{z}$ denotes the scattering coefficient in image patches.

For optimizing the problem in (8), there are difficulties that the iteration termination threshold in methods [11], [13] and the prior parameter in method [12], [27] are hard to be determined, which impacts the performance of scattering center extraction.

III. Proposed Method

Existing scattering center-based SAR target recognition methods first extract scattering centers via sparse optimization methods [11], [12], [13] and then design an appropriate classifier to gain promising performance under EOCs. The detached two stages may cause the mismatch of extracted scattering centers with classifier, which has impact on recognition results. In this article, by incorporating scattering center model into the deep learning framework, we build an EMI-Net for SAR target recognition under EOCs. In this section, we will present a comprehensive introduction to the structure of our EMI-Net, as well as its training and test procedures.

A. Overall Framework

The EMI-Net consists of two modules, and the first one is a feature extraction module that contains encoding and decoding processes and is utilized to extract the physical features, i.e., electromagnetic scattering centers, and the second one is a target recognition module that makes use of extracted scattering centers for recognition. The above two modules are embedded in an end-to-end framework, which is illustrated in Fig. 1. Fig. 1 shows that the original complex SAR images are fed into the scattering center extraction module to be encoded to obtain the physical features, which are then fed into the target recognition module to predict categories. Moreover, the extracted scattering center features are also decoded to obtain a sparse reconstructed image of the original SAR image. For an input SAR image $\mathbf{s}$ with size being $P \times P$, the pixel values of it are preprocessed by L2 normalization. The detailed architecture of the two modules in our model will be introduced as follows.

1) Scattering Center Extraction Module: According to the introduction in Section II, the scattering center extraction process can be seen as a sparse optimization problem. Thus, a sparse-solving algorithm is utilized to extract scattering centers based on the physical model of SAR images. Here, we adopt AMP [27] as the sparse-solving algorithm. As shown in Section II, given the input signal $\mathbf{s}$ and Fourier dictionary in the image domain $\mathbf{D}$, the iterative formula for the scattering coefficient $\mathbf{z}$ for the sparse solution of (8) is expressed as

$$\boldsymbol{\varepsilon}^{k-1} = \mathbf{s} - \mathbf{D}\mathbf{z}^{k-1} \tag{9}$$

$$\mathbf{z}^{k} = \xi_k\left(\mathbf{D}^T\boldsymbol{\varepsilon}^{k-1} + \mathbf{z}^{k-1}\right) \tag{10}$$

where $k \in \{1,\ldots,K\}$ denotes the iteration number with K denoting the total iteration number that is set according to the artificial experience, $\mathbf{D}^T$ represents the transpose of the dictionary $\mathbf{D}$, and $\xi_k(\cdot)$ denotes a nonlinear mapping function that is a soft threshold function [28]. Combining (9) with (10), the expression of (10) can be further derived as

$$\begin{aligned} \mathbf{D}^T\boldsymbol{\varepsilon}^{k-1} + \mathbf{z}^{k-1} &= \mathbf{D}^T\left(\mathbf{s}-\mathbf{D}\mathbf{z}^{k-1}\right) + \mathbf{z}^{k-1} \\ &\overset{\mathbf{s}=\mathbf{D}\bar{\mathbf{z}}}{=} \bar{\mathbf{z}} + \left(\mathbf{D}^T\mathbf{D}-\mathbf{I}\right)\left(\bar{\mathbf{z}}-\mathbf{z}^{k-1}\right). \end{aligned} \tag{11}$$

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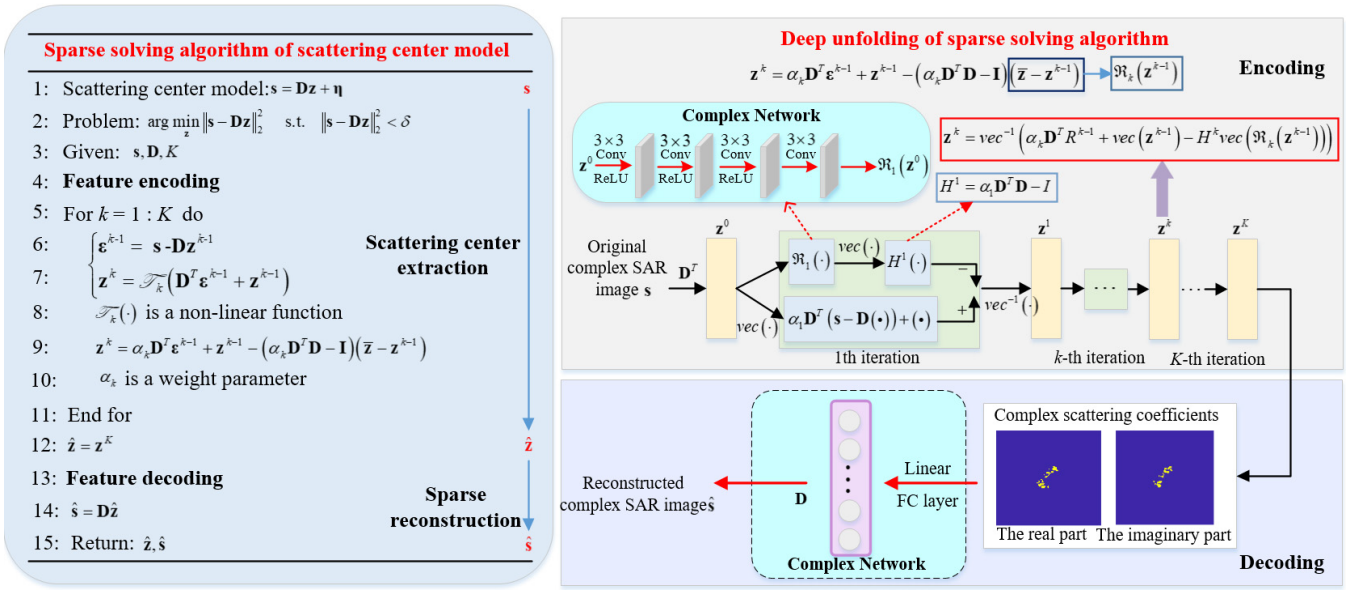


Fig. 2. Process of unfolding algorithm in our model. (a) AMP algorithm. (b) Architecture of the scattering center feature extraction module.

With $\bar{\mathbf{z}}$ denoting the expectation of the true value $\mathbf{z}$ and $(\mathbf{D}^T\mathbf{D}-\mathbf{I})(\bar{\mathbf{z}}-\mathbf{z}^{k-1})$ representing the noise term. Since the iteration process is a step-by-step process to approximate the true value, it is reasonable to replace the expectation $\bar{\mathbf{z}}$ in the first term of the right-hand side of (11) with the value $\mathbf{z}^k$ in kth iteration. Thus, (11) can be expanded as an iterative version and $\mathbf{z}^k$ is expressed as

$$\mathbf{z}^k = \mathbf{D}^T\boldsymbol{\varepsilon}^{k-1} + \mathbf{z}^{k-1} - \left(\mathbf{D}^T\mathbf{D}-\mathbf{I}\right)\left(\bar{\mathbf{z}}-\mathbf{z}^{k-1}\right). \tag{12}$$

To increase the flexibility of the AMP algorithm [27], a weight parameter α_k is introduced and (12) is rewritten as

$$\mathbf{z}^k = \alpha_k\mathbf{D}^T\boldsymbol{\varepsilon}^{k-1} + \mathbf{z}^{k-1} - \left(\alpha_k\mathbf{D}^T\mathbf{D}-\mathbf{I}\right)\left(\bar{\mathbf{z}}-\mathbf{z}^{k-1}\right). \tag{13}$$

With α_k denoting a trainable parameter. As shown in Fig. 2(a), we present the detailed steps of the AMP algorithm for scattering center extraction, consisting of feature encoding and decoding procedures. Since $\bar{\mathbf{z}}$ denotes the expectation of $\mathbf{z}$ that is obtained by the posterior inference. In real cases, the prior of $\mathbf{z}$ is needed to be set first, but it is hard to assume the accurate prior distribute of $\mathbf{z}$, and thus the estimation of $\bar{\mathbf{z}}$ is not accurate. In this way, the AMP algorithm is not flexible as well.

As indicated in the article [49], the unfolding algorithm is an effective way to replace the iteration solving of traditional algorithms with neural networks, which can avoid the difficulty of prior distribution settings. Moreover, some deep-unfolding networks [29], [48] have been proposed for tasks of compressive image sensing and anomaly detection. Inspired by this work, based on the scattering center model, we design a deep unfolding network for the AMP algorithm for scattering center extraction and SAR target recognition. In detail, a non-linear function is introduced in the AMP algorithm to fit the expectation of $\bar{\mathbf{z}}$. By the way of deep unfolding of the sparse solving algorithm, a non-linear mapping function $\Re_k(\cdot)$ is introduced to fit the term $(\bar{\mathbf{z}}-\mathbf{z}^{k-1})$, where $\Re_k(\cdot)$ is achieved by the convolution neural networks (CNNs) with multiple layers. Moreover, as discussed in Section II, the input image $\mathbf{s}$ is divided into m patches to reduce the computation complexity. Correspondingly, the scattering coefficient $\mathbf{z}$ is divided into m patches represented as $\sigma(\mathbf{z}, m)$. Thus, in (13), the scattering coefficient $\mathbf{z}^k$ in the kth iteration can be derived as

$$\mathbf{z}^k = \sigma^{-1}\left(\mathrm{vec}^{-1}\left(\begin{array}{c}\alpha_k\mathbf{D}^T R^{k-1} + \mathrm{vec}\left(\sigma\left(\mathbf{z}^{k-1}, m\right)\right) \\ -H^k vec\left(\Re_k\left(\sigma\left(\mathbf{z}^{k-1}, m\right)\right)\right)\end{array}\right), m\right) \tag{14}$$

$$H^k = \alpha_k\mathbf{D}^T\mathbf{D} - I \tag{15}$$

$$R^{k-1} = \mathbf{s} - \mathbf{D}\mathrm{vec}\left(\sigma\left(\mathbf{z}^{k-1}, m\right)\right) \tag{16}$$

where $\mathrm{vec}(\cdot)$ denotes the vectorization operation. Since the scattering coefficient $\mathbf{z}^k$ is complex-valued, the CNNs are constructed in a complex-valued neural network framework. For clear illustration, Fig. 2(b) shows the detailed architecture of the scattering center extraction module, which corresponds to the traditional AMP steps in Fig. 2(a). As presented in Fig. 2(b), K blocks of complex neural networks are constructed to unfold the K iterations in the AMP algorithm. Based on the deep learning framework, the unfolding network can avoid the issues of the prior setting, which is more flexible in practice and also has a better performance.

Moreover, in the below block of Fig. 2(b), the decoding process embeds the scattering center model into networks by setting the Fourier dictionary as the weight matrix of the linear layer, which achieves sparse decoding of SAR images. In detail, the scattering coefficient $\mathbf{z}$ is utilized to obtain the reconstructed image $\hat{\mathbf{s}}$ of the original SAR image. To embed the above process into neural networks, given the scattering coefficient $\mathbf{z}$, the reconstructed image $\hat{\mathbf{s}}$ can be generated by

$$\text{Latent layer: } \hat{\mathbf{s}} = \mathbf{D}\hat{\mathbf{z}} \tag{17}$$

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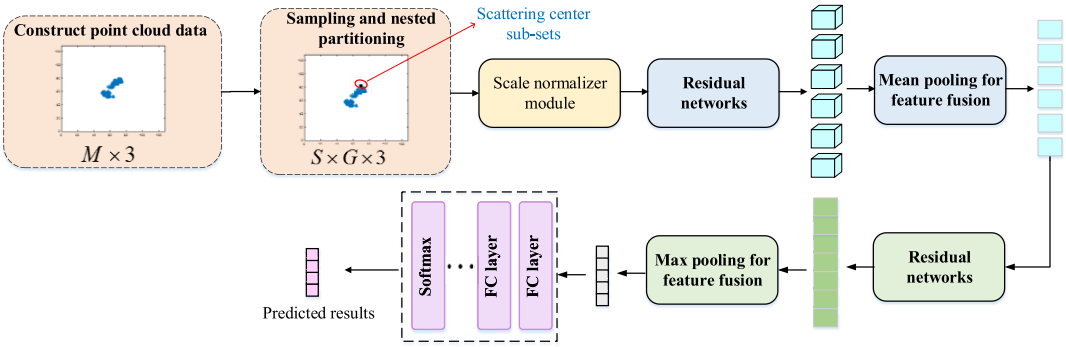


Fig. 3. Architecture of the target recognition module in our model.

where $\mathbf{D}$ denotes the weight matrix of a full connection layer and $\hat{\mathbf{z}}$ represents the estimated scattering coefficient $\mathbf{z}$. As presented in (17), the decoding process is implemented via a single liner full connection layer, whose weight matrix is fixed as Fourier matrix $\mathbf{D}$. In the image reconstruction process, the input–output and network parameter are complex-valued. Since the input are sparse scattering coefficients, the output image can remove the clutters and noises in the background area, and simultaneously remain the shape and pixels of targets.

Thus, the constructed scattering center extraction module is a mechanism-driven interpretable model that incorporates the physical model of SAR images into networks and proposes the deep unfolding network of a traditional sparse-solving algorithm. The output of this module is the scattering center comprising the locations and amplitude of dominant scattering centers in SAR targets, which reflect important physical characteristics of the target structure.

2) Target Recognition Module: To utilize the extracted scattering centers for SAR target recognition, we innovatively adopt a PCN as the classifier in our method. Since the scattering centers are a set of point data with each point composing of the 2-D locations (x- and y-axis) and 1-D amplitude, it is reasonable to adopt the PCN to deal with the scattering centers for classifying. Inspired by the PCN in papers [26], [47], we build a target recognition module with scattering centers for classifying, and the architecture of the target recognition module is illustrated in Fig. 3.

According to the flowchart in Fig. 3, we first construct 3-D point cloud data. Given M scattering centers, ith scattering center is represented as $C_i = (x_i, y_i, a_i)$ $(i = 1, \ldots, M)$ with (x_i, y_i) being 2-D locations and a_i amplitude. For an SAR image $\mathbf{s}$ with size being $P \times P$, the location of ith scattering center is represented as $(x_i, y_i) \in \{1, \ldots, P\}$, and its amplitude of ith scattering center is denoted as $a_i \in \{0, \ldots, 1\}$. In our manuscript, to deal with the problem of different scale between the location and amplitude, the max-normalization operation is adopted among the locations and amplitudes of M scattering centers in an image. Thus, the values of locations and amplitude are ranging from 0 to 1. For these M extracted scattering centers from an SAR image, the dimension of 3-D point cloud data is $M * 3$ $(i = 1, \ldots, M)$. Then the sampling and nested partitioning [25] are performed among the M points to obtain scattering center sub-sets with dimension being $S * G * 3$, and S denotes the number of sampling and G the number of points in each neighbor. Followed by the scale normalizer module, the data in each neighbor are processed into the same scale. In detail, for S neighbors with G scattering centers in each neighbor denoted as $\{C_{sg} = (x_{sg}, y_{sg}, a_{sg})\} \in \mathbb{R}^D$ with D being the dimension of a scattering center. The scale normalizer module transforms the local neighbor points by the following formulation:

$$\begin{aligned}\{C_{sg}\} &= \alpha \odot \frac{\{C_{sg}\} - C_g}{\sigma + \varepsilon} + \beta, \sigma \\ &= \sqrt{\frac{1}{D \times S \times G} \sum_{s=1}^{S} \sum_{g=1}^{G} (C_{sg} - C_g)}\end{aligned} \tag{18}$$

where $\alpha \in \mathbb{R}^D$ and $\beta \in \mathbb{R}^D$ are learnable parameters, $\odot$ indicates Hadamard production, and $\varepsilon = 1e^{-5}$ represents a small number for numerical stability. Thus, the scattering centers in each neighbor are normalized to the same magnitude. Then Residual networks are adopted to map the scaled data into an embedding space to obtain features with dimension $S \times G \times C'$, which can maintain the spatial relations of scattering centers. The feature fusion operation is performed to aggregation the information in each neighbor and output fused features with dimension $S * C'$ by a mean pooling layer. Then, other Residual networks are used to further extract the features with dimension being $S \times C_0$ for each sampling point, which further maps the features to higher dimensional vectors. With the maxpooling layer among S neighborhoods, the dimension of output is $1 \times C_0$. Finally, the fully connected layer is adopted to map the feature vector to the label and outputs the predicted results of each class. For briefly, this article neglects the introduction to techniques of sampling and nested partitioning, and the network architectures of Residual networks [25], [26].

As is known to all, traditional CNN utilizes the stripes and grain of input images as the key information for classifying, and thus the CNN classifier is very sensitive to the change of targets' scattering characteristics. In the cases of EOCs, the performance of the CNN classifier will degrade a lot. However, the PCN utilizes the physical features including the related locations and amplitude information of targets, and thus scattering centers are permutation-invariant data. Meanwhile,

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in the target recognition module, the Residual networks can learn representative features by mapping the point cloud data into an embedding space, and the operations of local region feature aggregation can fuse abundant information of local structure that depict the target structure characteristics.

3) Training and Test Procedures: As introduced in Section III-A, our model is built in an end-to-end framework that is composed of two modules, and we express its total loss function as

$$\text{Loss} = \sum_{i=1}^{N}\sum_{m=1}^{M}\left(R\left(\mathbf{s}_{im},\hat{\mathbf{s}}_{im}\right)+\lambda L(\mathbf{z}_{im})\right)+\sum_{i=1}^{N}H\left(\mathbf{y}_i,\hat{\mathbf{y}}_i\right) \quad (19)$$

where N means the number of training samples and M is the number of divided patches for each sample. $R(\mathbf{s}_{im},\hat{\mathbf{s}}_{im})$ denotes the reconstruction loss between the original sample $\mathbf{s}_{im}$ and the reconstructed sample $\hat{\mathbf{s}}_{im}$ from the mth patch in ith sample, which is further expanded as

$$R\left(\mathbf{s}_{im},\hat{\mathbf{s}}_{im}\right) = \|\mathbf{s}_{im}-\mathbf{D}\mathbf{z}_{im}\|^2 \quad (20)$$

$$\hat{\mathbf{s}}_{im} = \mathbf{D}\mathbf{z}_{im} \quad (21)$$

where $\|\cdot\|^2$ denotes the L2 normalization. Moreover, $L(\mathbf{z}_{im})$ denotes the sparse constraint for the scattering coefficient $\mathbf{z}_{im}$, and here we adopt the L1-norm to guarantee its sparsity. In addition, the parameter λ in (19) is an adjustable variable that is the trade-off between reconstruction loss and sparse constraint. Moreover, $H(\mathbf{y}_i,\hat{\mathbf{y}}_i)$ represents the cross-entropy loss between the true label $\mathbf{y}_i$ and predicted result $\hat{\mathbf{y}}_i$ for ith sample, and the loss is further expanded as

$$H\left(\mathbf{y}_i,\hat{\mathbf{y}}_i\right) = -\sum_{c=1}^{C} y_{ic}\log\left(\hat{y}_{ic}\right) \quad (22)$$

with C meaning the class number.

In the training procedure, the loss function in (19) is learned by the backward propagation algorithm [34] to joint optimize the network parameters in our model, including the parameter θ_e in scattering center extraction module and the parameter θ_r in target recognition module. Compared with some traditional scattering center extraction algorithms, our method is trained on a deep learning mechanism to attain better performance. For existing scattering center-based target recognition methods, the recognition process is independent with that of scattering center extraction, while our method is an end-to-end framework that avoids the mismatch between extracted scattering centers with classifier, and thus achieves better recognition performance.

In the test stage, the scattering center features of test samples can be directly extracted by feed-forward propagation via the trained scattering center extraction module with a parameter θ_e. Thus, the test procedure of our model has much higher time efficiency than traditional scattering center extraction algorithms, and then the recognition results can be predicted via the trained target recognition module with the parameter θ_r.

IV. Experiments

This section adopts the measured moving and stationary target acquisition and recognition (MSTAR) dataset to carry out some experiments to validate the performance of EMI-Net under SOC and EOCs. We first present the data description and experimental setup, and then give the results under SOC. The test images under EOC cases are simulated to verify the robustness of EMI-Net. Finally, we analyze our model's computation and space complexities.

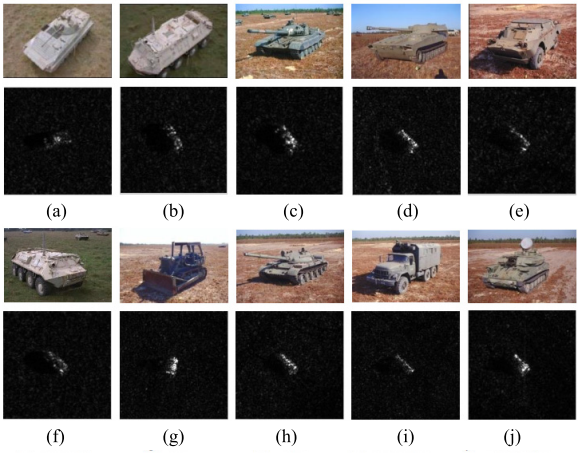


Fig. 4. Image examples of ten targets. The first row of (a)–(j) shows the optical images, and the second row of (a)–(j) shows corresponding SAR images. (a) BMP2, (b) BTR70, (c) T72, (d) 2S1, (e) BRDM2, (f) BTR60, (g) D7, (h) T62, (i) ZIL131, and (j) ZSU23/4.

A. Data Description

The MSTAR dataset is acquired by the Sandia National Laboratory in an X-band SAR sensor. In the dataset, each datum is a complex SAR image that is comprised of a real part and an imaginary part. The resolution and size of each image are 0.3×0.3 m and 128×128, respectively. The MSTAR dataset is taken as the benchmark with ten categories of cars, including BMP2, BTR70, T72, T62, BRDM2, BTR60, ZSU23/4, D7, ZIL131, and 2S1. For clear illustration, Fig. 4 shows some examples of optical images and SAR images in each category.

The operation conditions mainly include SOC and EOCs, and detailed data description is introduced in this section.

1) Data Description for SOC Case: During the data-collecting process in the SOC case, the operation conditions and the imaging parameters for training images are similar to those of test images. For the SAR images under SOC, targets' aspect angles range from 0° to 360°, while targets' depression angles for test and training images are respectively at 15° and 17°, which are slightly different from each other. Moreover, for each category of targets, the serial numbers of training images are the same as those of test images. Table I lists the serial number, depression, and sample number of the images in the SOC case.

2) Data Description for EOC Cases: In the EOCs, the imaging conditions of the training data are different from those of the test data. In the following, we consider four common EOC cases to verify the robust and efficient recognition performance of our model. The detailed description of the test data under four EOC cases is presented as follows.

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TABLE I
INFORMATION OF TRAINING AND TEST IMAGES IN SOC CASE

The category of target	Serial Number	Training images		Test images	
		Depression	Number	Depression	Number
BMP2	c21	17°	233	15°	195
BTR70	c71	17°	233	15°	196
T72	132	17°	232	15°	196
BTR60	7532	17°	256	15°	195
2S1	B01	17°	299	15°	274
BRDM2	E-71	17°	298	15°	274
D7	13015	17°	299	15°	274
T62	A51	17°	299	15°	273
ZIL131	E12	17°	299	15°	274
ZSU234	d08	17°	299	15°	274

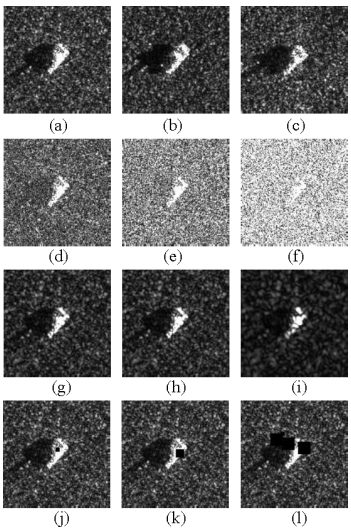


Fig. 5. SAR image samples of four EOC cases from T72 target. (a)–(c) are three versions of T72. (d)–(f) Show noised images for a sample from T72 with SNRs being 5, 0, and −5 dB, respectively. (g)–(i) Present the images with different resolutions for a sample from T72 with resolutions being 0.4 × 0.4 m, 0.5 × 0.5 m, and 0.6 × 0.6 m, respectively. (j)–(l) Show the images with occlusions, and the cover sizes are 5 × 5, 10 × 10, and 15 × 15, respectively.

a) EOC-1: EOC1-V, EOC1-C, and EOC1-D: The original MSTAR dataset contains three EOC cases that include EOC1-V (version variant), EOC1-C (configuration variance), and EOC1-D (depression angle variance), and the information on training and test images for the three types of EOC cases are respectively listed in Tables II–IV. For example, to give an intuitional illustration, Fig. 5(a)–(c) shows a sample with three versions of the T72 target. As we can see from Fig. 5(a)–(c), the three images belong to the same target but different versions, some local structures, such as fuel tank and battery, have slight difference from each other.

b) EOC-2: noise corruption: The MSTAR dataset is acquired under corporative conditions, and its SNR is high [23]. In real applications, there is noise interference in the process of collecting SAR data, including the noise from the radar system and background. To simulate the noise interference under the EOC-2 case, we manually add some complex Gaussian noises into the original complex SAR data. Fig. 5(d)–(f) respectively show the noised samples with SNRs being 5, 0, and −5 dB.

c) EOC-3: resolution variant: The resolution of images in the MSTAR dataset is 0.3 × 0.3 m. In the condition with different radar imaging parameters, such as signal bandwidth or the airplane flying speed, the resolution of images will change greatly. Thus, structure characteristics including the shape and outline of targets in SAR images will change. To simulate the images with different resolutions, we take the techniques in the work [21] to change the resolutions of test images from the original complex data in MSTAR dataset, and we present some image examples with different resolutions being 0.4 × 0.4 m, 0.5 × 0.5 m, and 0.6 × 0.6 m in Fig. 5(g)–(i).

d) EOC-4: occlusion: In real scenes, the interested targets may be covered by some occlusions. We simulate the random occlusions in the range of the target area for the test images in the MSTAR dataset. Since the size of the original image is 128 × 128, the target area is located in the central region 64 × 64 of the image. To test the generalization of our method, we simulate random occlusions in the central area with different sizes for test images. Fig. 5(j)–(l) respectively show examples with occluded sizes being 5 × 5, 10 × 10, and 15 × 15.

In our method, we set the size of the image patch as 32 × 32, and thus 16 patches are divided from an image. We adopt the Adam algorithm [35] as an optimizer with the learning rate being 2×10^{-4}. We set the weight parameter λ as 0.1. All experiments are performed under the platform of Pytorch framework with an NVIDIA GeForce RTX 3080 Ti GPU card and 128 GB of memory on the Ubuntu 18.04 Linux system.

B. Results Under SOC

In the SOC case, after the model training phase, the feature visualization and recognition results on test images are shown to comprehensively analyze our method under the SOC case.

1) Analysis of Our End-to-End Model: First of all, some experiments are conducted to evaluate the effect of end-to-end structure in our model on scattering center extraction compared with related methods, including SC-PCN, AMP [27], OMP [11], RELAX [36], and BCS [12]. Our EMI-Net and SC-PCN both obtain scattering coefficients by feed-forward propagation of the trained model. AMP infers the posterior distribution of scattering coefficients, whose iterative formulae are derived according to the message passing criterion. OMP searches the basis that is most aligned with the current residuals in the Fourier dictionary at the current iteration and the chosen basis is added to the set of selected bases, and then selected bases are utilized to calculate scattering coefficients. RELAX chooses the basis from the Fourier dictionary based on the criterion of maximum energy, and then it calculates scattering coefficients with the selected bases. BCS learns the posterior distribution of scattering coefficients with a variational Bayesian algorithm based on the sparse prior distribution assumption. In addition, hyper-parameters

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TABLE II
INFORMATION OF TRAINING AND TEST IMAGES IN EOC1-V (VERSION-VARIANT) CASE

Dataset	BMP2	BTR70	T72	BTR60	2S1	BRDM2	D7	T62	ZIL131	ZSU234
Training (17°)	233 (C21)	233	233 (132)	255	299	298	299	299	299	299
Test (15°)	587 (C21, 9566, 9563)	196	582 (132, S7, 812)	195	274	274	274	273	274	274

TABLE III
INFORMATION OF TRAINING AND TEST IMAGES IN EOC1-C (CONFIGURATION-VARIANT) CASE

Dataset	BMP2	BTR70	T72	BTR60	2S1	BRDM2	D7	T62	ZIL131	ZSU234
Training (17°)	233 (C21)	233	233 (132)	255	299	298	299	299	299	299
Test (15°)	292(9566, 9563)	196	386 (S7, 812)	195	274	274	274	273	274	274

TABLE IV
INFORMATION OF TRAINING AND TEST IMAGES IN EOC1-D (DEPRESSION-VARIANT) CASE

Dataset	Depression angle	2S1	BRDM2	ZSU234
Training	17°	299	298	299
Test	15°	274	274	274
	30°	288	287	288
	45°	303	303	303

TABLE V
RECOGNITION RESULTS OF DIFFERENT METHODS

Methods	SC-PCN			AMP	OMP	RELAX	BCS	
hyper-parameters	λ	ρ	m	γ	K	ε	M	φ
	0.5	10^{-5}	32	10^{-3}	150	10^{-3}	200	10^{-5}

of SC-PCN include the parameter λ that is the trade-off between reconstruction loss and sparse constraint, the size of image patches $m \times m$, and the learning rate ρ. The hyper-parameters of AMP [27], OMP [11], and RELAX [36] are the rate of change between two adjacent iterations γ, υ, and ε, respectively. The hyper-parameters of BCS [8] are the number of iterations M and prior parameter φ. Thus, Table V lists the hyper-parameter settings of SC-PCN, AMP [27], OMP [11], RELAX [36], and BCS [12].

The results of different methods are presented in Fig. 6. In detail, the iterative terminal threshold of OMP and RELAX is estimated by a method in [32], and the prior parameters in AMP and BCS are set based on the experiential values. We have presented the satisfying results for traditional methods in Fig. 6. As we can see from Fig. 6, the learned scattering coefficients of traditional methods are close to the results of our method, but they are still less precise in depicting targets' structure with some information missed. Since scattering center extraction effects of traditional methods are very sensitive to the setting of prior parameters or iteration termination threshold, there is a tradeoff to extracting scattering centers to keep a complete target structure and simultaneously avoid extracting noise points. Moreover, our model outperforms SC-PCN, since our model contains a recognition module to extract scattering centers that preserve more representative information. In particular, the results of our method are very close to those of the method without dividing patches, which illustrates that the operation of dividing patches will not impact the scattering center extraction performance. In our method, the original SAR images are divided as patches by way of overlapping sliding windows by utilizing the original image information as edge fills. Moreover, a Fourier dictionary is appropriately set for each image patch, and the scattering centers in an image patch can be extracted precisely based on the Fourier dictionary. Moreover, to quantitatively measure the scattering center extraction results of our method and related methods, we adopt two criteria including reconstruction error MSE and image entropy ENT, which are expressed as

$$\mathrm{MSE} = \sqrt{\frac{1}{N}\sum_{n=1}^{N}\frac{\left\|\hat{\mathbf{x}}_n - \mathbf{x}_n\right\|_2^2}{\left\|\mathbf{x}_n\right\|_2^2}} \tag{23}$$

$$\mathrm{ENT} = \frac{1}{N}\sum_{n=1}^{N}\left(-\left\|\frac{\mathbf{w}_n}{g}\right\|_2^2 \ln\left\|\frac{\mathbf{w}_n}{g}\right\|_2^2\right) \tag{24}$$

where $\mathbf{x}_n$ denotes nth original sample, $\hat{\mathbf{x}}_n$ denotes the nth reconstructed sample, N represents the number of samples, $\mathbf{w}_n$ denotes the learned scattering coefficient of the nth sample, $g = \|\mathbf{w}_n\|_2^2$ denotes the energy of $\mathbf{w}_n$, and $\|\cdot\|_2^2$ denotes the L2 normalization. In detail, MSE evaluate the difference between reconstructed images and original images. The small of the MSE is, the better the reconstruction effect is, which means that more detailed image information is preserved. ENT represents the degree of aggregation of extracted scattering centers. The small of the ENT is, the better the aggregation effect of extracted scattering centers is, which means that less discrete noise points are included. Thus, when the MSE and ENT are both smaller, the target region is retained better and simultaneously contains less discrete noise points, which illustrates the better effect of scattering center extraction. In Table VI, we list the MSE and ENT results for our method and related methods which shows that the results of our method is close to those of the method without dividing patches but both smaller than other methods. Thus,

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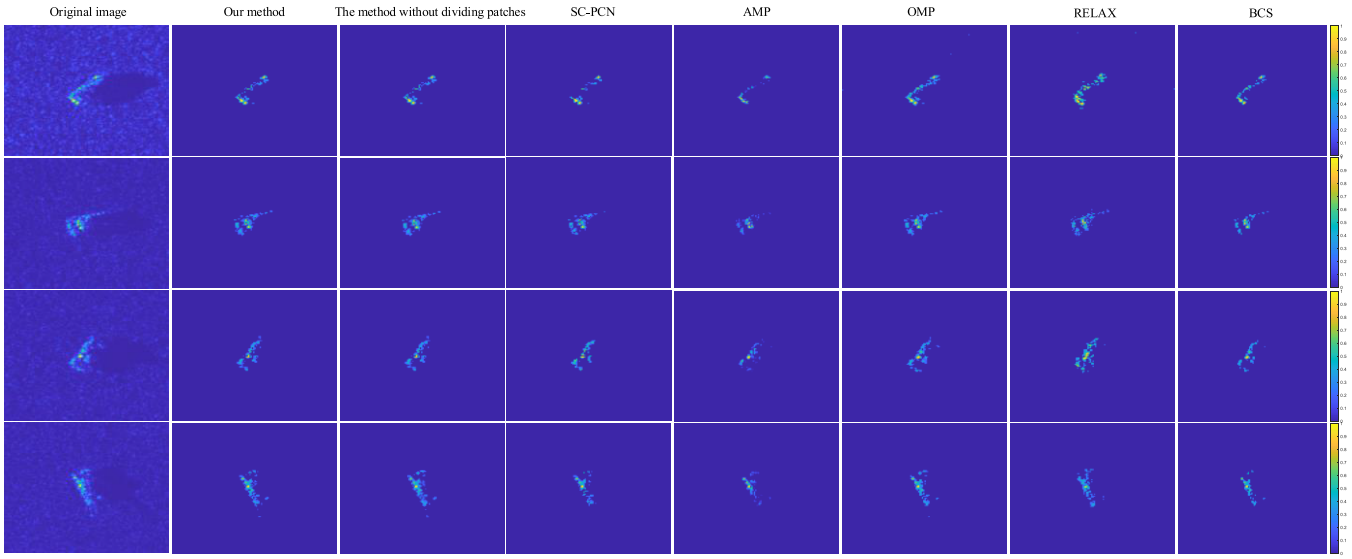


Fig. 6. Learned scattering coefficients by different methods for four SAR images samples. Each row shows the results of different methods for an SAR image, while each column presents the results of a method for four SAR images samples.

TABLE VI
RESULTS OF MSE AND ENT FOR DIFFERENT METHODS

Methods	EMI-Net	The method without dividing patches	SC-PCN	AMP	OMP	RELAX	BCS
MSE	0.3382	0.3331	0.3649	0.5114	0.4212	0.4105	0.4062
ENT	0.1317	0.1285	0.1383	0.1627	0.1465	0.1454	0.1473

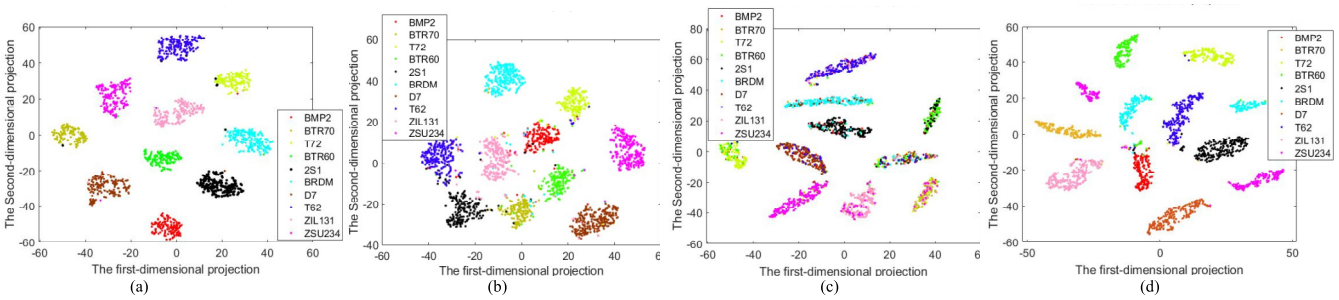


Fig. 7. t-SNE visualizations of features via different methods. (a) Our method, (b) SC-PCN, (c) OMP, and (d) VGG.

the qualitative and quantitative experiments validate that our method performs well as the method without dividing patches and achieves better scattering center extraction results than other methods.

Then, we visualize the separability of extracted features by our method and some related methods, including EMI-Net, SC-PCN, OMP [11], and VGG [15], and Fig. 7 shows the t-SNE results of features obtained by different methods. In Fig. 7, the features of each target gained by our method are more clearly separate than SC-PCN and OMP, which validates that our end-to-end model can improve feature separability. Moreover, since our method extracts physical features that contain more target structure information, the feature separability of our method outperforms VGG.

To further analyze the recognition performance of EMI-Net, the confusion matrix for each class is listed in Table VII, where each row shows the predicted recognition rates for the test samples from a specific class. The results in Table VII illustrate that our method can recognize each target well and gain promising average recognition results. We further analyze the recognition rate results of our EMI-Net compared with SC-PCN, OMP [11], and VGG [15], and the results are listed in Table VIII. In Table VIII, EMI-Net outperforms SC-PCN with 2.7% rates. In particular, SC-PCN is a two-stage method that is constructed to perform an ablation study to validate the effectiveness of our end-to-end mode, and it is not an existing method and cannot be given a reference. Moreover, since the SC-PCN [26], [29] is a two-stage version of our EMI-Net that has the same network settings and training parameter settings as EMI-Net, the performance improvement is only due to the end-to-end mode of our EMI-Net. In addition, EMI-Net also achieves better results than OMP and VGG, since EMI-Net learns the scattering center features based on a deep learning mode that contains useful information for recognition.

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TABLE VII
CONFUSION MATRIX OF OUR METHOD FOR TEST IMAGES FROM EACH TARGET UNDER SOC CASE

Targets	BMP2	BTR70	T72	BTR60	2S1	BRDM2	D7	T62	ZIL131	ZSU234
BMP2	98.96	0.53	0	0	0	0	0	0	0	0.51
BTR70	0	100.00	0	0	0	0	0	0	0	0
T72	0	0	100.00	0	0	0	0	0	0	0
BTR60	0	0	0	98.97	0	0	0	0	0	1.03
2S1	0	0	0.73	0	97.45	0.36	0	1.46	0	0
BRDM2	0	0	0	0	0	100.00	0	0	0	0
D7	0	0	0	0	0.16	0	99.68	0	0.16	0
T62	0	0	0	0	0	0	0	99.87	0.33	0
ZIL131	0	0	0	0	0	0	0.36	0	99.27	0.37
ZSU234	0	0	0	0	0	0	1.09	0	0	98.91
Average (%)	99.51									

TABLE VIII
RECOGNITION RESULTS OF DIFFERENT METHODS

Methods	**EMI-Net**	SC-PCN	OMP [11]	VGG [15]
PCC (%)	**99.51**	96.81	96.53	98.85

TABLE IX
RECOGNITION RESULTS OF DIFFERENT METHODS

Methods	**EMI-Net**	AMP -PCN	OMP -PCN	RELAX -PCN	BCS -PCN
PCC (%)	**99.51**	96.48	96.67	96.21	96.75

TABLE X
ABLATION STUDY RESULTS OF OUR METHOD

Classifier		End-to-end	PCC (%)
NN	PCN		
✓	×	×	96.65
✓	×	✓	98.80
×	✓	×	96.81 (SC-PCN)
×	✓	✓	**99.51 (EMI-Net)**

In addition, some ablation studies of our method have been conducted. First, we use some conventional methods in Fig. 6 including AMP [27], OMP [11], RELAX [36], and BCS [12] to extract scattering centers and input them into the designed PCN for target recognition, and the ablation studies results are listed in Table IX. As shown in Table IX, EMI-Net gains higher recognition rates than other methods, which shows that the scattering centers extracted by our method contain more representative information that are useful for recognition. Moreover, some relevant ablation studies also have performed to verify the advantage of PCN, and experimental results are presented in Table X. The results in Table X illustrates that the end-to-end mode and PCN in our model both devote to our promising performance. Since the scattering center features are discrete 3-D point data consisting of 2-D locations (x- and y-axis) and amplitude, the PCN is the network that is suitable for processing point data, while CNN is suitable to process image data with the structure information such as some points, lines, curves, etc. Therefore, PCN achieves better recognition results for scattering center features than CNN.

2) Comparison With Related Methods: To comprehensively compare the recognition performance of EMI-Net with some state-of-the-art methods, in Table XI, we list some recognition methods that are categorized into three types.

The first type of recognition method uses the amplitude of SAR images as the input. The sparse-representation-based classification (SRC) [37] and SVM [7] are respectively taken as the classifiers for the random forest (RF) features and PCA features. While the A-ConvNet [14], VGG [15], and MVGGNet [38] extract deep features for classifying. For the second type of method, attributed scattering centers (ASCs) and amplitude images are taken as the input, and then a feature fusion technique is further utilized to comprehensively fuse the scattering center features with RF features or deep features of amplitude images to gain superior recognition results. The third type of method takes complex SAR images as input, which extracts scattering center features and uses template matching techniques, deep networks, or PCNs as the classifier. Especially, the second type of method utilizes ASCs that contain more abundant target structure information than simple point scattering centers, and feature fusion of scattering centers and original images also contain much more useful information for recognition. Thus, they gain better recognition rates than other methods, especially ASC-DL gains the state art of performance.

In particular, as presented in Table XI, existing scattering center-based target recognition methods including FGL, BFM, SFM, OMP, and SC-PCN contain two independent stages, which may restrict their performance. Our method is an end-to-end model that achieves scattering center extraction and simultaneously targets recognition, which is devoted to promising recognition results. Therefore, our model achieves comparable performance to the second type of ASC-based methods even by extracting simple point scattering centers.

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TABLE XI
RECOGNITION PERFORMANCE OF DIFFERENT METHODS UNDER SOC CASE

Input	Methods	Features	End-to-end framework	Classifier	PCC (%)
Amplitude of SAR images	SRC [37]	RF features	No	SRC	96.29
	SVM [7]	PCA features	No	SVM	96.78
	A-ConvNet [14]	Deep features	Yes	CNN	99.13
	VGG [15]	Deep features	Yes	VGG16	98.85
	MVGGNet [28]	Deep features	Yes	CNN	99.27
Attributed scattering centers and amplitude of SAR images	FGL [21]	Attributed scattering centers and RF features	No	Template matching+SRC	99.08
	FEC [22]	Attributed scattering centers and deep features	No	Random forest	99.59
	ASC-DL [23]	**Attributed scattering centers and deep features**	**No**	**CNN**	**99.71**
	SDF-Net [47]	Attributed scattering centers and deep features	No	Point cloud network	99.58
Original complex SAR images	BFM [21]	Attributed scattering centers	No	Template matching	95.62
	SFM [38]	Attributed scattering centers	No	Template matching	95.24
	OMP	Scattering centers	No	VGG16	96.53
	SC-PCN	Scattering centers	No	Point cloud network	96.81
	Our method	Scattering centers	Yes	Point cloud network	99.51

TABLE XII
TEST PCCS OF OUR METHOD UNDER EOC1-V (VERSION-VARIANT) CASE

The category of target	Test images			
	Serial Number	Number	Each Type	Average
BMP2	9563	195	98.96	98.46
	c21	196	98.30	
	9566	196	98.16	
BTR70	c71	196	100.00	
T72	132	196	100.00	
	812	195	98.61	
	s7	191	97.35	
BTR60	7532	195	98.97	
2S1	B01	274	97.45	
BRDM2	E-71	274	100.00	
D7	13015	274	99.68	
T62	A51	273	99.87	
ZIL131	E12	274	99.27	
ZSU234	17°	274	98.91	

TABLE XIII
TEST PCCS OF OUR METHOD AND SOME COMPARISONS UNDER EOC1-V (VERSION-VARIANT) CASE

Methods	PCC (%)
PCA [43]	80.30
DL-JDSR [39]	91.48
AE [40]	87.04
DAE [41]	80.89
Euclidean AE [42]	91.29
VGGNet [37]	91.66
A-ConvNet [14]	92.19
OMP + VGGNet	92.65
CA-MCNN [17]	97.81
FGL [21]	98.46
FEC [22]	98.48
ASC-DL [23]	98.18
SDF-Net [47]	**98.76**
Our method	98.46

C. Results Under EOCs

We then examine the robust performance of our method under several EOCs for the test data from the MSTAR dataset.

1) Results Under EOC1 Cases: First of all, we examine the robust performance of our model when test data contain version variants. As listed in Table II, BMP2 and T72 targets respectively contain two version variants in their test data. In this subsection, we use our trained model to obtain the test results, and Table XII presents the test PCCs of our method on each type. In Table XII, test PCCs of the version variant are a little lower than those of the version non-variant. Due to the variation of shape and outline for BMP2 and T72 targets, their recognition results will degrade a little, but the average result only degrades about 1%, which shows the robust performance of our model under the EOC-1 case.

Moreover, Table XIII lists the recognition performance of our EMI-Net and some related methods. The deep methods are robust under the EOC-1 case and gain promising recognition results. As for FGL [21], FEC [22], ASC-DL [23], and SDF-Net [47], they fuse ASC features and deep features of images to achieve superior recognition results. Since ASCs depict SAR target characteristics more precisely than simple point scattering centers, these ASC-based methods slightly outperform EMI-Net. However, our method is an end-to-end model that avoids the mismatch between extracted features with classifier and learns discriminative information for recognition. Thus, as we can see for Table XIII, compared with ASC-based methods, our method still can obtain satisfying recognition result only with simple point scattering centers.

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TABLE XIV
TEST PCCS OF OUR METHOD AND SOME COMPARISONS UNDER EOC1-EOC1-C (CONFIGURATION VARIANCE) CASE

Methods	**Our method**	SVM [7]	SRC [45]	ASC [21]	CNN [14]	TJSR [44]	FGL [21]
PCC (%)	**93.32**	89.54	86.78	92.24	91.54	91.17	93.26

TABLE XV
TEST PCCS OF OUR METHOD AND SOME COMPARISONS UNDER EOC1-D (DEPRESSION ANGLE VARIANCE) CASE

Methods	PCC (%)		
	15°	30°	45°
SVM [7]	98.90	98.42	67.00
SRC [45]	98.42	94.79	56.66
ASC [21]	98.42	97.75	80.14
CNN [14]	99.12	97.42	64.17
TJSR [44]	98.94	95.24	70.73
FGL [21]	99.14	99.07	83.39
Our method	**99.51**	**99.34**	**83.52**

In addition, for the case of EOC1-C (configuration variance), the target class contains several different configurations, and the configurations in test data are different from those in training data. In this case, there are some local variations for the targets with the accessory or disassembly and so on. In Table XIV, we perform some experiments in this EOC1-C case, and the results show that our method still has satisfying performance and achieves better recognition rates than other methods.

Moreover, in the EOC1-D (depression angle variance) case, test images are obtained under quite different depression angles from those of training images. In this case, the training data are at 17°, while the test data are at 15°, 30°, and 45°. It is noting that the larger depression angle variance between training and test images is, the larger targets' structure variance exists. Table XV lists the experimental results of different methods under the EOC1-D case, which shows that the recognition rates of all of methods decrease when the depression angle variance increase. However, our method keeps robust performance and has much better recognition rates than related methods.

2) Results Under EOC-2 (Noise Corruption): The noise corruption for SAR images is a common case in real applications. Our method is capable of extracting scattering center features that are robust enough to depict SAR target structure information, such as the target shape, size, and so on. Since scattering centers are sparsely distributed in the targets, the reconstructed images with the extracted scattering centers can effectively remove the noise in the background. In Fig. 8, we show denoised images via our method and some traditional methods, and the five images in Fig. 8(a) from left to right are the SAR images under the SNRs being 10, 5, 0, −5, and −10 dB, respectively. The results in Fig. 8(b)–(e) are the reconstructed images obtained by our method, OMP, RELAX, and BCS algorithms, respectively. The results in Fig. 8(b) show that our method is effective on image denoising, even when the SNRs are low. As shown in Fig. 8(c)–(e), when the SNRs are higher than 5 dB, the OMP, RELAX, and BCS algorithms all perform well by removing most background noises and saving the target's shape; when the SNRs are between 0 and −10 dB, the OMP, RELAX, and BCS algorithms become invalid with lots of noises remained and targets illegible. In the OMP, RELAX, and BCS algorithms, the iteration numbers are hard to be estimated accurately, and thus their performance are restricted when test images are under low SNRs. Our model is based on deep learning *m* that is more flexible to extract scattering center and gains well-performing denoising results. Moreover, recognition results of EMI-Net are compared with those of OMP, RELAX, and BCS under different SNRs. In Table XVI, the results of OMP, RELAX, and BCS are obtained by using VGG16 as the classifier, and its input are reconstructed images. Table XVI verifies that our model gains much better results than OMP, RELAX, and BCS algorithms under different SNRs.

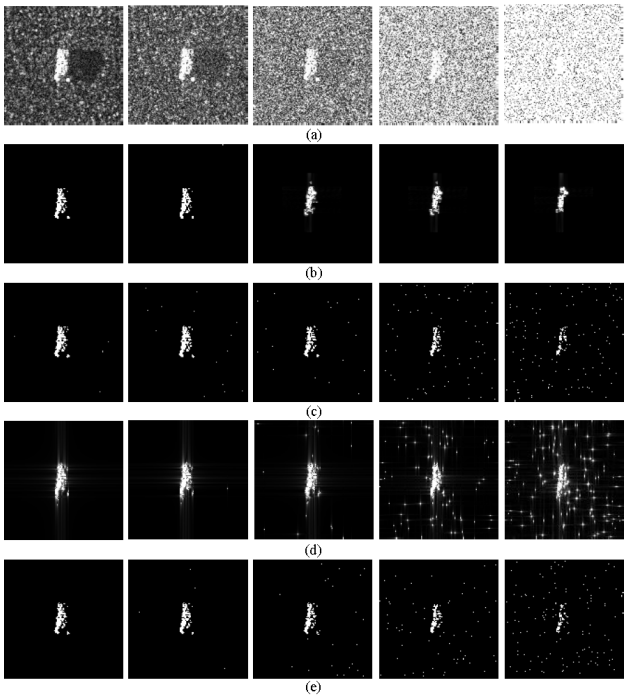


Fig. 8. Reconstructed images by different methods for SAR images under different SNRs. (a) Shows the noised SAR images under SNRs being 10, 5, 0, −5, and −10 dB and (b)–(e) show the reconstructed images by our method, OMP, RELAX, and BCS algorithms, respectively.

Furthermore, Fig. 9 presents the test recognition results of our EMI-Net method and some related comparison methods. Fig. 9 shows that the deep methods are more robust than traditional methods, in which the FGL, FEC, and BFM methods make use of the physical features to obtain fantastic robust results under low SNRs. Since the article [23] does not give the recognition results of ASC-DL under different SNRs, we cannot compare with ASC-DL under EOC-2. Since FGL, FEC, and BFM make use of ASCs that depict SAR targets more precisely and are more robust under low SNRs than

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TABLE XVI
TEST PCCS OF OUR METHOD AND SOME DENOISING METHODS UNDER DIFFERENT SNRS

Methods	PCC (%)				
	10dB	5dB	0dB	-5dB	-10dB
OMP	94.25	93.44	86.37	65.61	53.41
RELAX	93.18	91.28	86.12	61.37	51.03
BCS	95.67	94.32	88.53	66.32	54.16
Our method	**98.73**	**96.16**	**94.60**	**89.85**	**85.65**

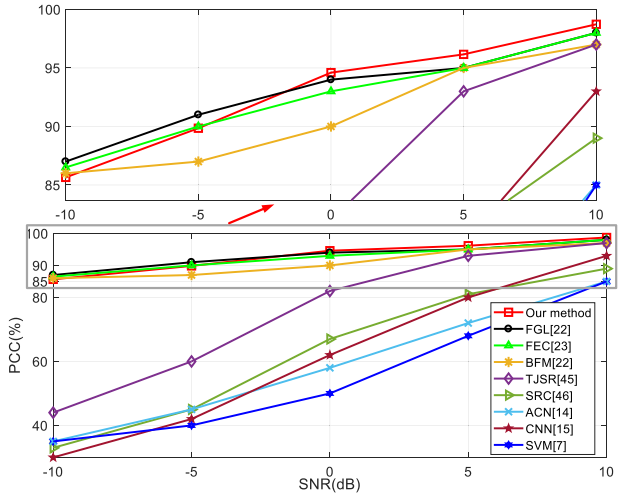


Fig. 9. Test PCCs of different methods under different SNRs (EOC-2).

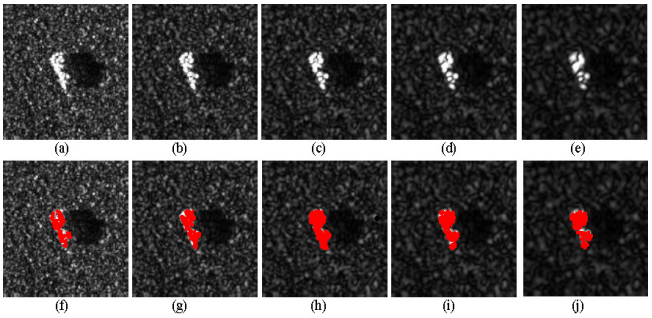


Fig. 10. Images with different resolutions and extracted scattering centers. (a)–(e) are the original images with resolutions being 0.3 × 0.3 m, 0.4 × 0.4 m, 0.5 × 0.5 m, 0.6 × 0.6 m, and 0.7 × 0.7 m, respectively, and (f)–(j) show the corresponding extracted scattering centers by our model.

point scattering centers, their results are a little better than our model when SNRs are lower than −5 dB. When SNRs are between 0 and 10 dB, our method gains better results, since it is an integrative model for extracting scattering centers and learning classifier that are beneficial to target recognition. Overall, Fig. 9 verifies robust recognition results of our method under EOC-2 case.

3) Results Under EOC-3 (Resolution Variant): To verify the robustness of our EMI-Net in the EOC-3 case, we decrease the resolutions of test images according to techniques in the article [4]. In the following, we decrease the resolutions of test images into 0.4 × 0.4 m, 0.5 × 0.5 m, 0.6 × 0.6 m, and 0.7 × 0.7 m, and each of image example is shown in

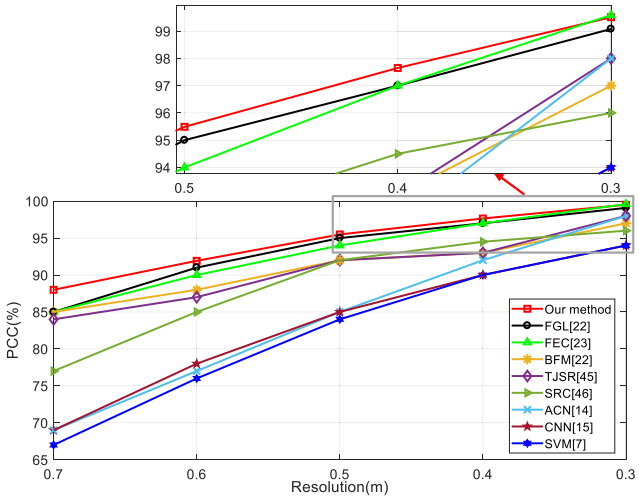


Fig. 11. Test PCCs of different methods for the test images under different resolutions (EOC-3).

TABLE XVII
TEST PCCS OF DIFFERENT METHODS EOC-4 CASE

Methods	PCC (%)			
	SOC	5×5	10×10	15×15
PCA	82.54	82.04	77.10	69.97
SVM	98.90	98.25	84.57	73.62
SRC	98.42	97.63	87.92	78.54
AE	88.73	87.89	81.53	74.36
A-ConvNet	94.84	94.31	86.97	77.32
VGG16	98.37	98.18	88.62	73.08
Our method	**99.51**	**99.23**	**95.74**	**84.67**

Fig. 10(b)–(e). As shown in Fig. 10(a)–(e), the lower the resolutions are, the more details of SAR targets decrease. Fig. 10(f)–(j) presents the corresponding extracted scattering centers by our model for the images in Fig. 10(a)–(e), which illustrates that extracted scattering centers can precisely locate the targets, even when the resolutions of test images are much lower than that of training images. Since our model is learned by the training data whose resolution is 0.3 × 0.3 m, the scattering centers of test data whose resolutions are much lower than 0.3 × 0.3 m still can be extracted precisely via the way of super-resolution with Fourier base. Thus, our model performs well on scattering center extraction under EOC-3.

To illustrate the robust recognition performance of our EMI-Net, Fig. 11 shows the PCCs of our EMI-Net and some comparisons when test images are under different resolutions, including 0.3 × 0.3 m, 0.4 × 0.4 m, 0.5 × 0.5 m, 0.6 × 0.6 m, and 0.7 × 0.7 m. Since ASC-DL [23] does not give the results under different resolutions, we cannot compare our method with the ASC-DL. Fig. 11 shows that our method outperforms other methods for the test images under different resolutions. In particular, when the resolutions of SAR images are lower than 0.6 × 0.6 m, the superiority of our model is more obvious. In the training phase, the Fourier dictionary being 0.3 × 0.3 m is fixed in our model, and thus our model learns the nonlinear mapping from input to the

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TABLE XVIII
COMPUTATION COMPLEXITY AND TIME COST (S) OF DIFFERENT METHODS IN TEST PROCEDURE

Input	Method	Computation complexity	Calculation (orders of magnitude)	Test time (seconds)
Amplitude of SAR images	PCA	$O\left(P^4+P^2\right)$	10^8	1.1647
	SAE	$O\left(\sum_{l=1}^{L} N_{in,l}N_{out,l}\right)+O\left(d^2\right)$	10^8	1.3374
	VGG16	$O\left(C_{\text{in},1}C_{\text{out},1}P^2K_1^2+\sum_{lc=2}^{LC-1} C_{\text{in},lc}C_{\text{out},lc}K_{lc}^2M_{lc}^2+\sum_{lf=1}^{LF} N_{in.lf}N_{out,lf}\right)$	10^8	1.2756
Attributed scattering centers and amplitude of SAR images	ASC-DL[24]	$O\left(K^2TR\right)+O\left(C_{\text{in},1}C_{\text{out},1}P^2K_1^2+\sum_{lc=2}^{LC-1} C_{\text{in},lc}C_{\text{out},lc}K_{lc}^2M_{lc}^2+\sum_{lf=1}^{LF} N_{in.lf}N_{out,lf}\right)$	10^{12}	-
Original complex SAR images	SC extraction by OMP + VGG16 classifier	$O\left(K^3\left(K+P^2\right)+P^2\right)+O\left(C_{\text{in},1}C_{\text{out},1}P^2K_1^2+\sum_{lc=2}^{LC-1} C_{\text{in},l}C_{\text{out},l}K_{lc}^2M_{lc}^2+\sum_{lf=1}^{LF} N_{in.lf}N_{out,lf}\right)$	10^{11}	161.0328
	The method without dividing patches	$O\left(3\left(C_{\text{in},1}C_{\text{out},1}P^2K_1^2+\sum_{lc=2}^{L-2} C_{\text{in},lc}C_{\text{out},lc}K_{lc}^2M_{lc}^2+P^4\right)\right)$	10^{11}	154.7417
	Our method	$O\left(3\left(C_{\text{in},1}C_{\text{out},1}p^2K_1^2+\sum_{lc=2}^{L-2} C_{\text{in},lc}C_{\text{out},lc}K_{lc}^2M_{lc}^2+p^4\right)\right)$	$\mathbf{10^6}$	**0.1284**

scattering centers with resolution being 0.3 × 0.3 m. In the test phase, when the resolution of test images is smaller than 0.3 × 0.3 m, our model still can extract the scattering centers of de-resolution test images via the way of super-resolution, which can obtain the scattering centers that are more precisely than those in the original de-resolution test images. Therefore, Fig. 11 validates that our model performs well on scattering center extraction under EOC-3.

4) Results Under EOC-4 (Occlusion): In the EOC-4 case, the test data with occlusions are introduced in Section IV-A. Since there is an occlusion in the target area, some details of targets are removed. In Table XVII, we list the test PCCs of our method and some other methods when the test images have different sizes of occlusions. We implement the results of some comparative methods in Table XVII. For the ASC-based methods including FGL [21], FEC [22], and ASC-DL [23], in the published papers, the occlusion conditions of [21] and [23] are different from our settings, and [22] do not conduct experiment under occlusion. Thus, our method cannot compare with FGL [21], FEC [22], and ASC-DL [23] under EOC-4.

In Table XVI, as the size of occlusions increases, the test PCCs of all methods gradually decrease. Compared with shallow models, deep models have better generalization ability and are more robust under EOC-4. For deep methods including AE, A-ConvNet, and VGG16, they rely on the shape and outline of targets for classifying. When the target area is occluded causing the targets' structure to be changed, the performance of the comparative methods will degrade a lot. Whereas our model extracts scattering centers that are robust under the EOC-4 case, it can attain more robust recognition results than other methods.

D. Computation and Space Complexities

The computation and space complexities of recognition methods reflect their practicability in real applications.

In Tables XVIII and XIX, we respectively analyze the computation and space complexities of different methods, and here we choose some representative methods in each type. The first type of methods includes the PCA, SAE, and VGG16 that use the amplitude of SAR images as input. ASC-DL [23] belongs to the second type of methods that first extracts ASCs and then adopts a network for classifying with the ASCs and amplitude of SAR images. For the third type of methods, the OMP first extracts the scattering centers of complex images and then obtains the sparse reconstructed images, which are fed into the VGG16 classifier. Moreover, our model and the method without dividing patches takes complex SAR images as input.

Since PCA and OMP both contain two independent stages of feature extraction and classifying, their computation and space complexities contain two parts. The size of the original SAR images is $P \times P$. In detail, the PCA extracts the features with dimension d from SAR images, and the SVM [22] is utilized for classifying. For SAE with L fully connected layers, $N_{\text{in},l}$ and $N_{\text{out},l}$ respectively denote the dimension of input and output in lth layer. In particular, the dimension of input in the 1st layer is P^2, and the dimension of output in the Lth layer is the same as the number of classes. After feature extraction by SAE, the SVM is used as the classifier. VGG16 contains LC convolution layers and LF fully connected layers, where $C_{\text{in},lc}$ and $C_{\text{out},lc}$ represent the channel numbers in the lcth input and output layer, K_{lc} and M_{lc} denote the sizes of feature maps in the lcth input and output layer, $N_{\text{in}.lf}$ and $N_{\text{out},lf}$ represent the similar meaning as those in SAE. Compared with

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TABLE XIX
SPACE COMPLEXITY AND MEMORY REQUIREMENT (BYTE) OF DIFFERENT METHODS IN TEST PROCEDURE

Input	Method	Space complexity	Memory requirement (bytes)
Amplitude of SAR images	PCA	$O\left(P^4\right)$	2.68×10^8
	SAE	$O\left(\sum_{l=1}^{L} N_{in,l}N_{out,l} + P^4\right)$	2.72×10^8
	VGG16	$O\left(\sum_{lc=1}^{L} C_{\text{in},lc}C_{\text{out},lc}K_{lc}^2 + \sum_{lc=1}^{L-1} C_{\text{out},lc}M_{lc}^2 + C_{\text{out,L}}P^2 + \sum_{lf=1}^{LF} N_{in.lf}N_{out,lf}\right)$	1.67×10^8
Attributed scattering centers and amplitude of SAR images	ASC-DL[24]	$O\left(7K^2M_EN_EG\right) + O\left(\sum_{lc=1}^{L} C_{\text{in},lc}C_{\text{out},lc}K_{lc}^2 + \sum_{lc=1}^{L-1} C_{\text{out},lc}M_{lc}^2 + C_{\text{out,L}}P^2 + \sum_{lf=1}^{LF} N_{in.lf}N_{out,lf}\right)$	1.29×10^9
Original complex SAR images	SC extraction by OMP + VGG16 classifier	$O\left(2\left(P^4+P^2\right)\right) + O\left(2\left(\sum_{lc=1}^{L} C_{\text{in},lc}C_{\text{out},lc}K_{lc}^2 + \sum_{lc=1}^{L-1} C_{\text{out},lc}M_{lc}^2 + C_{\text{out,L}}P^2\right) + \sum_{lf=1}^{LF} N_{in.lf}N_{out,lf}\right)$	5.43×10^8
	The method without dividing patches	$O\left(2\left(\sum_{lc=1}^{LC-1} C_{\text{in},lc}C_{\text{out},lc}K_{lc}^2 + P^4 + \sum_{lc=1}^{LC-1} C_{\text{out},lc}M_{lc}^2 + C_{\text{out,L}}P^2\right) + \sum_{lf=1}^{LF} N_{in.lf}N_{out,lf}\right)$	5.39×10^8
	Our method	$O\left(2\left(\sum_{lc=1}^{L-1} C_{\text{in},lc}C_{\text{out},lc}K_{lc}^2 + p^4 + \sum_{lc=1}^{L-1} C_{\text{out},lc}M_{lc}^2 + C_{\text{out,L}}p^2\right) + \sum_{lf=1}^{LF} N_{in.lf}N_{out,lf}\right)$	$\mathbf{2.16\times10^6}$

the VGG16 classifier, OMP needs to extract scattering centers first with K being the number of iterations, and K is usually close to P. Moreover, the ASC-DL [23] utilize the genetic algorithm [46] to extract ASCs, where $M_E \times N_E$ means the size of sparse dictionary and G denotes the dictionary grid number. To compare the complexities fairly, we assume that VGG classifiers have the same structure. Since the ASCs have more parameters than SCs, the process of ASCs extraction has much higher computation complexity. In our method, the size of input patches p is usually much smaller than P, and our model contains V convolution layers and M fully connected layers that are much smaller than those in the VGG16. The orders of magnitude of calculation in second column of Table XVIII indicates lower computation complexity. In the last column of Table XIX, we show the test time of some methods. As listed in Table XVIII, the time cost of SAE is lower than that of PCA, since SAE infers test results by feed-forward propagation. The VGG16 contains multiple convolution layers, and its time cost is much higher than that of SAE. Since OMP is time-consuming in the test stage, whose time cost is much higher than that of the VGG16 classifier. Since the codes of ASC-DL are not public online and we cannot implement the codes by ourselves, we cannot give its test time. In our model, the numbers of convolution and fully connected layers are smaller than those of VGG16 and the dimension of input patch is far smaller than that of VGG16, our model costs less time in test stage and shows much higher time-efficiency.

Furthermore, Table XIX lists the space complexity and memory cost of different methods. As we can see from Table XIX, our model attains a much lower memory requirement. To sum up, on the consideration of time cost and memory requirement, our model gains much higher time-efficiency and simultaneously lower space cost, which validates the practicability of our model.

V. CONCLUSION

Inspired by the excellent robustness of scattering centers, this article focuses on constructing a novel deep model that extracts scattering center features for SAR target recognition under EOCs. Our model integrates scattering center extraction and target recognition in an end-to-end framework, which can avoid the mismatch between scattering center features and classifier. Moreover, we incorporate the scattering center model into deep networks and build a mechanism-driven interpretable model to learn physical features of SAR images, which learns scattering center features precisely on the deep learning mode. Based on image-domain scattering center model, we adopt the technique of dividing input SAR images into multiple patches to reduce computation and space complexities of our model. In addition, since the scattering centers are composed of 3-D data, i.e., the locations (x- and y-axis) and amplitude, we adopt a PCN as the classifier with scattering centers for target recognition. Based on the measured MSTAR dataset, the experimental results validate that EMI-Net gains promising scattering center extraction and target recognition performance under EOCs, and has high time-efficiency and low space requirement.

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